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// 13_array_input_display.cpp
// Accept integers from the user and display all elements of the array

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;

    int arr[n];

    cout << "Enter " << n << " integers:\n";
    for (int i = 0; i < n; i++) {
        cout << "Element [" << i << "]: ";
        cin >> arr[i];
    }

    cout << "\nArray elements: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i];
        if (i < n - 1) cout << ", ";
    }
    cout << endl;

    return 0;
}
```

```
PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 13_array_input_display.cpp -o 13_array_input_display } ;
Enter the number of elements: 5
Enter 5 integers:
Element [0]: 10
Element [1]: 20
Element [2]: 30
Element [3]: 40
Element [4]: 50

Array elements: 10, 20, 30, 40, 50
```

```
// 14_largest_smallest.cpp
// Find the largest and the smallest element in an array

#include <iostream>
using namespace std;

int main() {
```

```

int n;
cout << "Enter the number of elements: ";
cin >> n;

int arr[n];

cout << "Enter " << n << " integers:\n";
for (int i = 0; i < n; i++) {
    cout << "Element [" << i << "]: ";
    cin >> arr[i];
}

int largest = arr[0], smallest = arr[0];

for (int i = 1; i < n; i++) {
    if (arr[i] > largest) largest = arr[i];
    if (arr[i] < smallest) smallest = arr[i];
}

cout << "\nLargest element : " << largest << endl;
cout << "Smallest element : " << smallest << endl;

return 0;
}

```

```

PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 14_largest_smallest.cpp -o 14_largest_smallest } ; if ($?) {
Enter the number of elements: 6
Enter 6 integers:
Element [0]: 45
Element [1]: 12
Element [2]: 78
Element [3]: 23
Element [4]: 56
Element [5]: 9

Largest element : 78
Smallest element : 9

```

```

// 15_sum_average.cpp
// Calculate the sum and average of all elements in an array

```

```

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;

```

```

int arr[n];

cout << "Enter " << n << " integers:\n";
for (int i = 0; i < n; i++) {
    cout << "Element [" << i << "]: ";
    cin >> arr[i];
}

int sum = 0;
for (int i = 0; i < n; i++) {
    sum += arr[i];
}

double average = (double)sum / n;

cout << "\nSum : " << sum << endl;
cout << "Average : " << average << endl;

return 0;
}

```

```

PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 15_sum_average.cpp -o 15_sum_average } ; if ($?) { .\15_sum_average.exe
Enter the number of elements: 5
Enter 5 integers:
Element [0]: 10
Element [1]: 20
Element [2]: 30
Element [3]: 40
Element [4]: 50

Sum      : 150
Average  : 30

```

```

// 16_even_odd_count.cpp
// Count the number of even and odd elements in an array

```

```

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;

    int arr[n];
    cout << "Enter " << n << " integers:\n";
}

```

```

for (int i = 0; i < n; i++) {
    cout << "Element [" << i << "]: ";
    cin >> arr[i];
}

int evenCount = 0, oddCount = 0;

for (int i = 0; i < n; i++) {
    if (arr[i] % 2 == 0)
        evenCount++;
    else
        oddCount++;
}

cout << "\nCount of Even elements : " << evenCount <<
endl; cout << "Count of Odd elements : " << oddCount <<
endl;

return 0;
}

```

```

PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 16_even_odd_count.cpp -o 16_even_odd_count } ; if ($?) {
Enter the number of elements: 6
Enter 6 integers:
Element [0]: 12
Element [1]: 7
Element [2]: 24
Element [3]: 15
Element [4]: 8
Element [5]: 9

Count of Even elements : 3
Count of Odd elements : 3

```

```

// 17_linear_search.cpp
// Search for a given element using Linear Search and display its position

```

```

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;

    int arr[n];
    cout << "Enter " << n << " integers:\n";

```

```

for (int i = 0; i < n; i++) {
    cout << "Element [" << i << "]: ";
    cin >> arr[i];
}

int key;
cout << "\nEnter the element to search: ";
cin >> key;

bool found = false;
for (int i = 0; i < n; i++) {
    if (arr[i] == key) {
        cout << "Element " << key << " found at position " << (i + 1) << "
(index " << i << ")." << endl;
        found = true;
    }
}

if (!found)
    cout << "Element " << key << " not found in the array." << endl;

return 0;
}

```

```

PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 17_linear_search.cpp -o 17_linear_search } ; if ($?) { .\17_linear_search.exe
Enter the number of elements: 6
Enter 6 integers:
Element [0]: 12
Element [1]: 7
Element [2]: 24
Element [3]: 15
Element [4]: 8
Element [5]: 9

Enter the element to search: 24
Element 24 found at position 3 (index 2).

```

```

// 18_reverse_array.cpp
// Display the elements of an array in reverse order

```

```

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;
}

```

```

int arr[n];

cout << "Enter " << n << " integers:\n";
for (int i = 0; i < n; i++) {
    cout << "Element [" << i << "]: ";
    cin >> arr[i];
}

cout << "\nOriginal array : ";
for (int i = 0; i < n; i++) {
    cout << arr[i];
    if (i < n - 1) cout << ", ";
}

cout << "\nReversed array : ";
for (int i = n - 1; i >= 0; i--) {
    cout << arr[i];
    if (i > 0) cout << ", ";
}

cout << endl;

return 0;
}

```

```

PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 18_reverse_array.cpp -o 18_reverse_array } ; if ($?) { .\18_reverse_array.exe
Enter the number of elements: 6
Enter 6 integers:
Element [0]: 12
Element [1]: 7
Element [2]: 24
Element [3]: 15
Element [4]: 8
Element [5]: 9

Original array : 12, 7, 24, 15, 8, 9
Reversed array : 9, 8, 15, 24, 7, 12

```

```

// 19_bubble_sort.cpp
// Sort the elements of an array in ascending order using Bubble Sort

```

```

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;
}

```

```

int arr[n];

cout << "Enter " << n << " integers:\n";
for (int i = 0; i < n; i++) {
    cout << "Element [" << i << "]: ";
    cin >> arr[i];
}

// Bubble Sort
for (int i = 0; i < n - 1; i++) {
    bool swapped = false;
    for (int j = 0; j < n - i - 1; j++) {
        if (arr[j] > arr[j + 1]) {
            int temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
            swapped = true;
        }
    }
    // If no swaps occurred, array is already sorted
    if (!swapped) break;
}

cout << "\nSorted array (Ascending): ";
for (int i = 0; i < n; i++) {
    cout << arr[i];
    if (i < n - 1) cout << ", ";
}
cout << endl;

return 0;
}

```

```

PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 19_bubble_sort.cpp -o 19_bubble_sort } ; if ($?) { .\19_bi
Enter the number of elements: 6
Enter 6 integers:
Element [0]: 45
Element [1]: 12
Element [2]: 78
Element [3]: 23
Element [4]: 56
Element [5]: 9

Sorted array (Ascending): 9, 12, 23, 45, 56, 78

```

```

// 20_merge_arrays.cpp
// Merge two arrays into a third array and display the merged array

```

```

#include <iostream>
using namespace std;

int main() {
    int m, n;

    cout << "Enter the number of elements in Array 1: ";
    cin >> m;
    int arr1[m];
    cout << "Enter " << m << " integers for Array 1:\n";
    for (int i = 0; i < m; i++) {
        cout << "Element [" << i << "]: ";
        cin >> arr1[i];
    }

    cout << "\nEnter the number of elements in Array 2: ";
    cin >> n;
    int arr2[n];
    cout << "Enter " << n << " integers for Array 2:\n";
    for (int i = 0; i < n; i++) {
        cout << "Element [" << i << "]: ";
        cin >> arr2[i];
    }

    int merged[m + n];

    // Copy first array
    for (int i = 0; i < m; i++)
        merged[i] = arr1[i];

    // Copy second array
    for (int i = 0; i < n; i++)
        merged[m + i] = arr2[i];
    cout << "\nMerged array: ";
    for (int i = 0; i < m + n; i++) {
        cout << merged[i];
        if (i < m + n - 1) cout << ", ";
    }
    cout << endl;

    return 0;
}

```



```

PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 20_merge_arrays.cpp -o 20_merge_arrays } ; if ($?) { .\20
Enter the number of elements in Array 1: 4
Enter 4 integers for Array 1:
Element [0]: 10
Element [1]: 20
Element [2]: 30
Element [3]: 40

Enter the number of elements in Array 2: 4
Enter 4 integers for Array 2:
Element [0]: 50
Element [1]: 60
Element [2]: 70
Element [3]: 80

Merged array: 10, 20, 30, 40, 50, 60, 70, 80

```

```
// 21_remove_duplicates.cpp
```

```
// Remove duplicate elements from an array and display only unique elements
```

```

#include <iostream>
using namespace std;

```

```

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;

    int arr[n];

    cout << "Enter " << n << " integers:\n";
    for (int i = 0; i < n; i++) {
        cout << "Element [" << i << "]: ";
        cin >> arr[i];
    }
    int unique[n];
    int uniqueCount = 0;

    for (int i = 0; i < n; i++) {
        bool isDuplicate = false;
        for (int j = 0; j < uniqueCount; j++) {
            if (arr[i] == unique[j]) {
                isDuplicate = true;
                break;
            }
        }
        if (!isDuplicate) {

```

```

        unique[uniqueCount++] = arr[i];
    }
}

cout << "\nOriginal array : ";
for (int i = 0; i < n; i++) {
    cout << arr[i];
    if (i < n - 1) cout << ", ";
}

cout << "\nUnique elements : ";
for (int i = 0; i < uniqueCount; i++) {
    cout << unique[i];
    if (i < uniqueCount - 1) cout << ", ";
}
cout << endl;

return 0;
}

```

```

PS E:\MIHARKA> cd "e:\MIHARKA" ; if ($?) { g++ 21_remove_duplicates.cpp -o 21_remove_duplicates } ; if ($?) {
Enter the number of elements: 7
Enter 7 integers:
Element [0]: 10
Element [1]: 20
Element [2]: 10
Element [3]: 30
Element [4]: 20
Element [5]: 40
Element [6]: 30

Original array : 10, 20, 10, 30, 20, 40, 30
Unique elements : 10, 20, 30, 40
}

// 22_left_rotation.cpp
// Perform a left rotation of an array by one position

```

```

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter the number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter " << n << " integers:\n";
    for (int i = 0; i < n; i++) {
        cout << "Element [" << i << "]: ";
    }
}

```

```

        cin >> arr[i];
    }
    cout << "\nOriginal array : ";
    for (int i = 0; i < n; i++) {
        cout << arr[i];
        if (i < n - 1) cout << ", ";
    }
    int first = arr[0];
    for (int i = 0; i < n - 1; i++) {
        arr[i] = arr[i + 1];
    }
    arr[n - 1] = first;
    cout << "\nAfter left rotation : ";
    for (int i = 0; i < n; i++) {
        cout << arr[i];
        if (i < n - 1) cout << ", ";
    }
    cout << endl;
    return 0;
}

```

```

PS E:\NIHARKA> cd "e:\NIHARKA" ; if ($?) { g++ 22_left_rotation.cpp -o 22_left_rotation } ; if ($?) { .\;
Enter the number of elements: 6
Enter 6 integers:
Element [0]: 10
Element [1]: 20
Element [2]: 30
Element [3]: 40
Element [4]: 50
Element [5]: 60

Original array      : 10, 20, 30, 40, 50, 60
After left rotation : 20, 30, 40, 50, 60, 10

```